

Drone Programming Assignment

In week 3 of 2026, spring semester in LAMK – Instructor Petri Hannula assigned students in robotics R504D101-3003 to utilize a programmable drone through the obstacle course in the LAMK basement. This report consists of the second team of students to complete the assignment. This team initially consisted of four members, but in 21.1 the group added members to the group who did not attend the first lecture.

The task utilized a DJI Tello drone and the assignment included connecting to the drone via wifi. After this, students were expected to run a python file which contained instructions for the drone to maneuver through the obstacle course. An instruction library was available, namely, the `` djitellopy ``python module. Along with the teachers instructions was also a book which contained some API instructions to move the drone, specifically moving commands.

The overall completion time for this task was roughly six hours over three meetings (1.21, 1.16, 1.15) The deliverables include this report which contains the code, pictures, and a link to a video.

[Link to Google Drive Video](#)

Difficulties

The group had expected to only perform this task in a couple hours, but the group had unexpected problems with the task. During flight testing and runs, the drone would unexpectedly land, drift, and make unexpected moves. The solution was found via a Teams message to Petri Hannula, who explained the problem and showed an instruction video which showed how to solve it. The issue was umi calibration error and the solution was to calibrate the drone before continuing the task.

The group also had some minor tasks retrieving fully charged batteries to complete the task at the beginning.

Reflection

Upon the report writer's reflection, the assignment perhaps could have been performed a bit more smoothly had the initial step to the instructions of the task had been to calibrate the drone.

In addition, the group's inexperience using drones was a primary difficulty in achieving the task, though still completely doable.

Attachments



Picture 1 – The dji tello drone on its designated starting position.



Picture 2 – The obstacle course.

```
from djitellopy import Tello
import time

# initialization
tello = Tello()
tello.connect()

# Takeoff
tello.takeoff()

# Obstacle 1
tello.move_up(70)
tello.move_forward(500)
tello.move_forward(160)
# Obstacle 2
tello.move_down(110)
tello.move_right(440)

# Obstacle 3
tello.move_up(110)
tello.move_back(170)
tello.move_down(40)
# Obstacle 5 -- Zigzag
tello.move_left(70)

# Obstacle 6 -- Zigzag
tello.move_back(110)
tello.move_right(140)
# Obstacle 7 -- Zigzag
tello.move_back(100)

# Landing
tello.move_left(100)
tello.move_back(200)
tello.land()

# closing connection
tello.end()
```

Picture 3 – The Python code used to achieve the result.

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